

PREDICTION OF LUNG CANCER USING LOGISTIC REGRESSION ANALYSIS

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Certified that this is a bonafied report of project work entitled

***“PREDICTION OF LUNG CANCER USING
LOGISTIC REGRESSION ANALYSIS”***

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*Submitted in partial fulfillment of requirements for the course work in IV
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This is to certify that the project work entitled

***“PREDICTION OF LUNG CANCER USING
LOGISTIC REGRESSION ANALYSIS”***

Carried by

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during IV Semesters of their study under my guidance.

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DECLARATION

We declare that the work presented in this project has been jointly carried out by us and I have done the same work in any form of previously. I have done this project work as a part of M.Sc. Statistics course.

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CHAPTER-1

INTRODUCTION TO LUNG CANCER

1. DEFINITION OF CANCER

Cancer is a disease in which cells in the body grow out of control and is one of the most serious health problems in the world. Among various different types of cancer, lung cancer is one of the leading causes of death in both men and women.

Lung cancer was uncommon before the advent of cigarette smoking; it was not even recognized as a distinct disease until 1761. Different aspects of lung cancer were described further in 1810. Malignant lung tumours made up only 1% of all cancers seen at autopsy in 1878, but had risen to 10-15% by the early 1900s. Case reports in the medical literature numbered only 374 worldwide in 1912, but a review of autopsies showed the incidence of lung cancer had increased from 0.3% in 1852 to 5.66% in 1952. In Germany in 1929, physician Fritz Lickint recognized the link between smoking and lung cancer, which led to an aggressive antismoking campaign. The British Doctors' Study, published in the 1950s, was the first solid epidemiological evidence of the link between lung cancer and smoking. As a result, in 1964 the Surgeon General of the United States recommended smokers should stop smoking.

The first successful pneumonectomy for lung cancer was performed in 1933. Palliative radiotherapy has been used since the 1940s. Radical radiotherapy, initially used in the 1950s, was an attempt to use larger radiation doses in patients with relatively early-stage lung cancer, but who were otherwise unfit for surgery. In 1997, CHART was seen as an improvement over conventional radical radiotherapy. With SCLC, initial attempts in the 1960s at surgical resection and radical radiotherapy were unsuccessful. In the 1970s, successful chemotherapy regimens were developed.

1.1 TYPES OF LUNG CANCER

Lung cancer is divided into 2 main types: Non-small cell lung cancer (NSCLC) and small cell lung cancer (SCLC). These types grow and spread differently. They are often treated in different ways.

Non-small cell lung cancer (NSCLC):

About 85% to 90% of lung cancers are non-small cell. This cancer has 3 major types. They are grouped by the kind of lung cell the cancer started in and by how the cells look under a microscope. They have slight differences among them. But they tend to have a similar outlook (prognosis) and are generally treated the same way.

- **Adenocarcinoma.** This is the most common type of NSCLC. It's the most common type of lung cancer in non smokers. But it's found more often in smokers or former smokers. It tends to grow in the outer edges of the lungs. It usually grows more slowly than other types of lung cancer.
- **Squamous cell carcinoma (epidermoid carcinoma).** This type of NSCLC develops more often in smokers or former smokers. These cancers tend to start in the middle part of the lungs near the main airways (the bronchi).
- **Large cell carcinoma.** This is the least common type of NSCLC. It tends to quickly grow and spread to other organs. This can make it harder to treat.

Small cell lung cancer (SCLC):

Only about 1 in 10 to 3 in 20 people with lung cancer have small cell lung cancer. It is also called oat cell cancer. It is almost only found in smokers. It grows and spreads more quickly than non-small cell lung cancer. It often spreads to other parts of the body at an early stage. Lung cancer, like all cancers, can act differently in each person, depending on the kind of lung cancer it is and the stage it is in. But when lung cancer spreads outside the lungs, it often goes to the same places. The first-place lung cancer usually spreads to is the lymph nodes in the center of the chest. These lymph nodes are called mediastinal lymph nodes.

Lung cancer may also spread to the lymph nodes in the lower neck. In its later stages, lung cancer may spread (metastasize) to distant parts of the body, like the liver, brain, or bones. Lung cancer is one of the most common and deadliest forms of cancer globally, responsible for more deaths than breast, colon, and prostate cancers combined. It is characterized by the uncontrolled growth of abnormal cells in the lung tissues, which can spread to other parts of the body through metastasis. Despite significant advancements in medical technology and cancer research, lung cancer remains a critical public health challenge due to its aggressive nature, late-stage diagnosis, and complex treatment options. This chapter provides a comprehensive overview of lung cancer, including its causes, risk factors, symptoms, diagnostic methods, and the impact of modern technologies on its detection and management.

1.2 Overview of Lung Cancer

Lung cancer primarily originates in the cells lining the bronchi and other parts of the lung. There are two main types: non-small cell lung cancer (NSCLC), which accounts for about 85% of all cases, and small cell lung cancer (SCLC), which is more aggressive but less common. NSCLC includes subtypes like adenocarcinoma, squamous cell carcinoma, and large cell carcinoma. SCLC, on the other hand, tends to spread more quickly and is often diagnosed at an advanced stage.

1.3 Global Impact and Prevalence

According to the World Health Organization (WHO), lung cancer is the leading cause of cancer-related deaths worldwide, with over 2.2 million new cases and 1.8 million deaths annually. This high mortality rate is largely due to late detection, as the disease often presents with subtle symptoms that are easily overlooked or mistaken for other respiratory conditions. In regions with high smoking rates and poor air quality, the incidence of lung cancer is significantly higher, highlighting the need for better awareness and early screening programs.

2.Causes and Risk Factors

The primary cause of lung cancer is smoking, responsible for nearly 85% of all cases. However, non-smokers are also at risk due to factors like air pollution, genetic mutations, and exposure to harmful chemicals such as asbestos and radon. Early detection is crucial, as it dramatically improves survival rates, but this is often difficult because symptoms can remain hidden until the disease reaches an advanced stage.

The primary cause of lung cancer is long-term exposure to carcinogens, with tobacco smoke being the leading contributor, responsible for approximately 85% of all cases. However, non-smokers are also at risk due to factors like air pollution, exposure to radon gas, occupational hazards (e.g., asbestos, arsenic), and genetic predisposition. Studies have shown that secondhand smoke, diesel exhaust, and industrial chemicals can also significantly increase lung cancer risk. Recent genetic research has identified specific mutations, like those in the EGFR, KRAS, and ALK genes, which can drive the development of lung cancer even in non-smokers.

2.1 Early Symptoms and Diagnosis

Early symptoms of lung cancer are often mild and nonspecific, including a persistent cough, chest pain, shortness of breath, hoarseness, unexplained weight loss, and coughing up blood (hemoptysis). As the disease progresses, symptoms may become more severe and include difficulty swallowing, bone pain, and neurological problems if the cancer spreads to the brain. Diagnostic methods include imaging tests like chest X-rays, CT scans, PET scans, and biopsies. Molecular testing can also identify genetic mutations, helping to guide targeted therapy decisions.

2.2 Challenges in Early Detection

Lung cancer is one of the most serious and widespread forms of cancer, accounting for a significant portion of global cancer-related deaths. It arises from the uncontrolled growth of abnormal cells in the lung tissues, which can spread to other parts of the body if left untreated. Despite major advancements in cancer research, lung cancer remains a critical challenge due to its aggressive nature, late-stage diagnosis, and complex treatment options.

One of the biggest challenges in lung cancer treatment is early detection. Most lung cancers are diagnosed at an advanced stage, when the prognosis is poor. This is partly due to the lungs' large size and capacity to compensate for impaired function, which allows tumors to grow undetected for months or even years. Additionally, the lack of effective screening programs in many countries further complicates early diagnosis.

3.Advancements in Technology and Treatment

Recent advancements in medical imaging, artificial intelligence (AI), and precision medicine have transformed lung cancer diagnosis and treatment. AI-based image analysis, for example, has shown promise in identifying lung nodules and predicting cancer risk with greater accuracy. Targeted therapies, immunotherapy, and personalized medicine are also improving survival rates by tailoring treatment to the genetic makeup of individual tumors.

3.1 Importance of Prevention and Awareness

Given the high mortality rate of lung cancer, prevention and early detection are critical. Public health campaigns focusing on smoking cessation, reducing exposure to air pollution, and promoting regular health check-ups can significantly reduce the incidence of lung cancer. Additionally, raising awareness about early symptoms and risk factors can encourage at-risk populations to seek medical advice sooner, potentially improving survival rates.

3.2 Symptoms

Common symptoms of lung cancer include:

1. Persistent Coughing: A cough that doesn't go away or gets worse over time.
2. Chest Pain or Discomfort: Pain or discomfort in the chest, shoulder, or back.
3. Coughing up Blood or Rust-Colored Mucus: Coughing up blood or rust-colored mucus can be a sign of lung cancer.
4. Shortness of Breath: Feeling short of breath or experiencing wheezing.
5. Fatigue: Feeling tired or weak.
6. Weight Loss: Unexplained weight loss.

3.3 Diagnosis

Diagnosing lung cancer typically involves:

1. Imaging Tests: X-rays, CT scans, MRI scans, or PET scans can help detect tumors.
2. Biopsies: A biopsy involves removing a sample of tissue from the lungs for examination.

3.4 Treatment Options

Treatment options for lung cancer depend on the stage and type of cancer, as well as overall health. Options may include:

1. Surgery: Surgery can be used to remove tumors or affected lung tissue.
2. Chemotherapy: Chemotherapy uses medications to kill cancer cells.
3. Radiation Therapy: Radiation therapy uses high-energy rays to kill cancer cells.
4. Targeted Therapy: Targeted therapy uses medications that target specific cancer cells.
5. Immunotherapy: Immunotherapy uses medications that stimulate the immune system to fight cancer.

3.5 Prevention

Preventing lung cancer involves:

1. Quitting Smoking: Quitting smoking can significantly reduce the risk of lung cancer.
2. Avoiding Exposure to Carcinogens: Avoiding exposure to radon, asbestos, and other carcinogens can reduce the risk.
3. Maintaining a Healthy Lifestyle: Eating a healthy diet, exercising regularly, and maintaining a healthy weight can help reduce the risk.

4. Conclusion

In summary, lung cancer is a complex and challenging disease with high mortality rates, primarily due to its aggressive nature and late-stage diagnosis. However, advancements in technology, personalized medicine, and early detection hold promise for improving patient outcomes. Continued research, public awareness, and early screening programs are essential in the global fight against lung cancer.

Using the patient's various cardiac characteristics and the R language approach of logistic regression on a publicly accessible dataset from Kaggle, we developed and examined models for predicting lung cancer disease in this project.

This study aims to establish the most significant heart disease risk factors and estimate the overall risk using logistic regression.

To overcome difficulties with classification, we employ logistic regression. Instead of using linear regression, which denotes continuous progress, it does this by forecasting categorical outcomes. An example of a binomial, which has two possible results in the most straightforward instance, is the prediction of heart disease, which has been steadily rising in prevalence worldwide. When compared to the current method, using logistic regression increases Accuracy.

The primary goal of developing this approach was to forecast the likelihood of developing heart disease ten years from now. To train our system, we used a variety of feature selection strategies, including backward elimination and logistic regression. The details of these algorithms are covered below. The Kaggle dataset, which has 800 observations, forecasts the likelihood that a patient in a specific location will develop heart disease. In this study, we used R software to predict heart disease using the patient data provided. Over the next ten years, with the aid of this patient data, heart disease development can be anticipated.

CHAPTER-2

METHODOLOGY

1. Study Design

The study adopts a quantitative, observational research design to develop a predictive model for lung cancer diagnosis using logistic regression, implemented in the R programming language. Logistic regression is a well-established statistical method ideally suited for binary classification tasks, such as predicting the presence or absence of lung cancer (1 = cancer, 0 = no cancer). Its strengths include interpretability, computational efficiency, and the ability to model relationships between a binary outcome and multiple predictor variables, making it a preferred choice in medical research.

The observational nature of the study leverages secondary data from a publicly available dataset or clinical database, ensuring ethical data collection and accessibility for research purposes. This approach eliminates the need for primary data collection, reducing time and resource constraints while maintaining data quality. The data is processed, analyzed, and modeled using R, a powerful open-source statistical computing environment widely used for data analysis, visualization, and machine learning. R's extensive libraries facilitate robust statistical modeling and visualization, enabling comprehensive exploration and evaluation of the predictive model.

The study aims to contribute to early lung cancer detection by identifying significant predictors and developing a reliable model that can assist clinicians in risk assessment. The methodology is structured to ensure reproducibility, transparency, and scientific rigor, with each step carefully documented to support future validation or extension of the work.

2.LOGISTIC REGRESSION

Logistic regression is a type of regression analysis in statistics used for prediction of outcome of a categorical dependent variable from a set of predictor or independent variables. In logistic regression the dependent variable is always binary. Logistic regression is mainly used to for prediction and also calculating the probability of success. Logistic regression (also known as **Binomial logistics regression**) in R programming is a classification

algorithm used to find the probability of event success and event failure. It is used when the dependent variable is binary (0/1, True/False, Yes/No) in nature.

In other words, **Logistic regression** is a **supervised machine learning algorithm** used for **classification tasks** where the goal is to predict the probability that an instance belongs to a given class or not. Logistic regression is a statistical algorithm which analyse the relationship between two data factors. The article explores the fundamentals of logistic regression, it's types and implementations.

Logistic regression is used for binary classification where we use sigmoid function, that takes input as independent variables and produces a probability value between 0 and 1.

Sigmoid Function

- The sigmoid function is a mathematical function used to map the predicted values to probabilities.
- It maps any real value into another value within a range of 0 and 1. The value of the logistic regression must be between 0 and 1, which cannot go beyond this limit, so it forms a curve like the “S” form.
- The S-form curve is called the Sigmoid function or the logistic function.
- In logistic regression, we use the concept of the threshold value, which defines the probability of either 0 or 1. Such as values above the threshold value tends to 1, and a value below the threshold values tends to 0.

Now we use the sigmoid function where the input will be z and we find the probability between 0 and 1. i.e. predicted y .

$$\sigma(z) = \frac{1}{(1+e^{-z})}$$

sigmoid function converts the continuous variable data into the probability i.e. between 0 and 1.

- $\sigma(z)$ tends towards 1 as $z \rightarrow \infty$

- $\sigma(z)$ tends towards 0 as $z \rightarrow -\infty$
- $\sigma(z)$ is always bounded between 0 and 1

where the probability of being a class can be measured as:

$$P(y=1)=\sigma(z)$$

$$P(y=0)=1-\sigma(z)$$

For example, we have two classes Class 0 and Class 1 if the value of the logistic function for an input is greater than 0.5 (threshold value) then it belongs to Class 1 otherwise it belongs to Class 0. It's referred to as regression because it is the extension of linear regression but is mainly used for classification problems.

Mathematical Implementation

Logistic regression is a type of **generalized linear model (GLM)** used for classification tasks, particularly when the response variable is binary. The goal is to model the probability that a given input belongs to a particular category. The output represents a probability, ranging between 0 and 1. It can be expressed as:

$$P = 1 / (1 + e^{-(\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n)})$$

In logistic regression, the **odds** represent the ratio of the probability of success to the probability of failure. The **odds ratio (OR)** is a key concept that helps interpret logistic regression coefficients. It measures how the odds change with a one-unit increase in a predictor variable:

- An OR of **1** indicates equal probability of success and failure.
- An OR of **2** means success is twice as likely as failure.
- An OR of **0.5** implies failure is twice as likely as success.

$$\text{Odds} = \frac{P}{1 - P}$$

Since the outcome is binary and follows a **binomial distribution**, logistic regression uses the **logit link function**, which connects the linear model to the probability:

$$\text{logit}(P) = \log\left(\frac{P}{1-P}\right) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n$$

This transformation ensures that the predicted probabilities stay within the (0, 1) interval and that the model is linear in the log-odds.

Logistic regression estimates the model parameters using **maximum likelihood estimation**. This approach finds the coefficients that make the observed outcomes most probable. Each coefficient β_i in the logistic regression model represents the change in the **log-odds** of the outcome for a one-unit increase in the corresponding predictor x_i , assuming all other variables are held constant.

- If $\beta_i > 0$, an increase in x_i increases the probability of success.
- If $\beta_i < 0$, an increase in x_i decreases the probability of success.

This interpretation is crucial when analyzing how predictor variables influence the predicted outcome.

Key Points:

- Logistic regression predicts the output of a categorical dependent variable. Therefore, the outcome must be a categorical or discrete value.
- It can be either Yes or No, 0 or 1, true or False, etc. but instead of giving the exact value as 0 and 1, it gives the probabilistic values which lie between 0 and 1.
- In Logistic regression, instead of fitting a regression line, we fit an “S” shaped logistic function, which predicts two maximum values (0 or 1).

2.1 Types of Logistic Regression

On the basis of the categories, Logistic Regression can be classified into three types:

1. **Binomial:** In binomial Logistic regression, there can be only two possible types of the dependent variables, such as 0 or 1, Pass or Fail, etc.

2. Multinomial: In multinomial Logistic regression, there can be 3 or more possible unordered types of the dependent variable, such as “cat”, “dogs”, or “sheep”
3. Ordinal: In ordinal Logistic regression, there can be 3 or more possible ordered types of dependent variables, such as “low”, “Medium”, or “High”.

Assumptions of Logistic Regression

We will explore the assumptions of logistic regression as understanding these assumptions is important to ensure that we are using appropriate application of the model. The assumption include:

1. Independent observations: Each observation is independent of the other. meaning there is no correlation between any input variables.
2. Binary dependent variables: It takes the assumption that the dependent variable must be binary or dichotomous, meaning it can take only two values. For more than two categories SoftMax functions are used.
3. Linearity relationship between independent variables and log odds: The relationship between the independent variables and the log odds of the dependent variable should be linear.
4. No outliers: There should be no outliers in the dataset.
5. Large sample size: The sample size is sufficiently large

2.2How does Logistic Regression work?

The logistic regression model transforms the linear regression function continuous value output into categorical value output using a sigmoid function, which maps any real-valued set of independent variables input into a value between 0 and 1. This function is known as the logistic function.

Let the independent input features be:

$$X = \begin{bmatrix} x_{11} & \cdots & x_{1m} \\ \vdots & \ddots & \vdots \\ x_{n1} & \cdots & x_{nm} \end{bmatrix}$$

and the dependent variable is Y having only binary value i.e. 0 or 1.

$$Y = \begin{cases} 0 & \text{if class 1} \\ 1 & \text{if class 2} \end{cases}$$

then, apply the multi-linear function to the input variables X

$$z = \sum_{i=1}^n (w_i x_i) + b$$

Here x_i is the i th observation of X, $w_i = [w_1, w_2, w_3, \dots, w_m]$ is the weights or Coefficient, and b is the bias term also known as intercept. simply this can be represented as the dot product of weight and bias.

$$z = w \cdot X + b$$

Equation of Logistic Regression:

The odd is the ratio of something occurring to something not occurring. it is different from probability as the probability is the ratio of something occurring to everything that could possibly occur. so odd will be:

$$\begin{aligned} \Rightarrow p(X) &= \frac{e^{(\beta_0 + \beta_1 x)}}{e^{(\beta_0 + \beta_1 x)} + 1} \\ \Rightarrow p(e^{(\beta_0 + \beta_1 x)} + 1) &= e^{(\beta_0 + \beta_1 x)} \\ \Rightarrow p \cdot e^{(\beta_0 + \beta_1 x)} + p &= e^{(\beta_0 + \beta_1 x)} \\ \Rightarrow p &= e^{(\beta_0 + \beta_1 x)} - p \cdot e^{(\beta_0 + \beta_1 x)} \\ \Rightarrow p &= e^{(\beta_0 + \beta_1 x)} (1 - p) \\ \Rightarrow \frac{p}{1 - p} &= e^{(\beta_0 + \beta_1 x)} \\ \Rightarrow \ln\left(\frac{p}{1 - p}\right) &= \beta_0 + \beta_1 x \end{aligned}$$

At the core of logistic regression is the **logistic (or sigmoid) function**, which maps any real valued input to a value between 0 and 1 which is interpreted as a probability. This allows the model to describe the relationship between the input features and the probability of the binary outcome.

2.3 Terminologies involved in Logistic Regression

Here are some common terms involved in logistic regression:

- **Independent variables:** The input characteristics or predictor factors applied to the dependent variable's predictions.
- **Dependent variable:** The target variable in a logistic regression model, which we are trying to predict.
- **Logistic function:** The formula used to represent how the independent and dependent variables relate to one another. The logistic function transforms the input variables into a probability value between 0 and 1, which represents the likelihood of the dependent variable being 1 or 0.
- **Odds:** It is the ratio of something occurring to something not occurring. It is different from probability as the probability is the ratio of something occurring to everything that could possibly occur.
- **Log-odds:** The log-odds, also known as the logit function, is the natural logarithm of the odds. In logistic regression, the log odds of the dependent variable are modelled as a linear combination of the independent variables and the intercept.
- **Coefficient:** The logistic regression model's estimated parameters, show how the independent and dependent variables relate to one another.
- **Intercept:** A constant term in the logistic regression model, which represents the log odds when all independent variables are equal to zero.
- **Maximum likelihood estimation:** The method used to estimate the coefficients of the logistic regression model, which maximizes the likelihood of observing the data given the model

2.4 Properties of logit model

1. As P goes from 0 to 1 (i.e., as Z varies from $-\infty$ to $+\infty$), the logit L goes from $-\infty$ to $+\infty$.
2. Although L is linear in X , the probabilities themselves are not. This property is in contrast with the LPM model where the probabilities increase linearly with X .
3. Although we have included only a single X variable, or regressor, in the preceding model, one can add as many regressors as may be dictated by the underlying theory.
4. The logit becomes negative and increasingly large in magnitude as the odds ratio decreases from 1 to 0 and becomes increasingly large and positive as the odds ratio increases from 1 to infinity.
5. The interpretation of the logit model is as follows: β_2 , the slope, measures the change in L for a unit change in X , that is, it tells how the log-odds in favor of owning a house change as income changes by a unit, say, Rs. 1000. The intercept β_1 is the value of the log-odds in favor of owning a house if income is zero. Like most interpretations of intercepts, this interpretation may not have any physical meaning.
6. Whereas the LPM assumes that P_i is linearly related to X_i , the logit model assumes that the log of the odds ratio is linearly related to X_i .

CHAPTER-3 INSTRUMENTS

1.SOFTWARE DESCRIPTION

R is a programming language and free software developed by Ross Ihaka and Robert Gentleman in 1993. R possesses an in-depth catalogue of applied mathematics and graphical strategies. It includes Machine Learning in R language algorithms, simple and linear regression, statistics, applied mathematics. Most of the R libraries are written in R, except for serious machine tasks, C, C++, and algebraic language codes are most well-liked.

R is not solely entrusted by academics, however many massive firms and MNC's additionally use R programming language, including Uber, Google, Airbnb, Facebook, and then on. Data analysis with R is finished in an exceedingly series of steps: programming, transforming, discovering, modelling, and communicating the results.

1.1 EVOLUTION OF R

R was initially written by Ross Ihaka and Robert Gentleman at the Department of Statistics of the University of Auckland in Auckland, New Zealand. R made it's first appearance in 1993.

A large group of individuals has contributed to R by sending code and bug reports.

Since mid-1997 there has been a core group (the "R Core Team") who can modify the R source code archive.

2.R VERSION

R is free software and comes with ABSOLUTELY NO WARRANTY. You are welcome to redistribute it under certain conditions. Type 'license()' or 'licence()' for distribution details. Natural language support but running in an English locale

R is a collaborative project with many contributors. Type 'contributors()' for more information and 'citation()' on how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or 'help.start()' for an HTML browser interface to help. Type 'q()' to quit R.

2.1 FEATURES OF R

As stated earlier, R is a programming language and software environment for statistical analysis, graphics representation and reporting. The following are the important features of R

- R is a well-developed, simple and effective programming language which includes conditionals, loops, user defined recursive functions and input and output facilities.
- R has an effective data handling and storage facility,
- R provides a suite of operators for calculations on arrays, lists, vectors and matrices.
- R provides a large, coherent and integrated collection of tools for data analysis.
- R provides graphical facilities for data analysis and display either directly at the computer or printing at the papers.

2.2 USAGE OF R PROGRAMMING

R language is in much demand in real-world applications because of the following reasons:

IMPORTANT FOR DATA SCIENCE: As R is an interpreted language, we can run code without any compiler which is most important in data science. R is a vector language and hence powerful and faster than another language. R is used in biology, genetics as well as in statistics. Hence, it can perform any type of task.

OPEN-SOURCE: R language is an open-source language. It is also maintained by a large number of the programmers as a community across the world. Since R is issued under the General Public License (GNU), and hence there is no restriction on its usage.

POPULARITY: R programming language has become the most popular programming language in the technological world. R language is not given importance in the academic world but with the emergence of data science, the requirement for R in industries has increased.

ROBUST VISUALIZATION LIBRARY: R language consists of libraries like ggplot2, plot that provides graphical plots to the user. R is mostly recognized for its amazing visualizations which is very important in data science programming language.

USED TO DEVELOP WEB APPS: R provides the ability to build web applications. Using the R package, we can create develop interactive applications using the console of your R IDE.

PLATFORM INDEPENDENT: R language is a platform-independent language. It can work on any system irrespective of whether it is Windows, Linux, and Mac.

USED IN MACHINE LEARNING IN R LANGUAGE: Most important advantage of R programming is that it helps to carry out Machine Learning in R language operations like classification, regression and also provides features for artificial intelligence and neural networks.

3.RSTUDIO APPLICATION FOR WINDOWS

RStudio is free, open-source integrated development environment for R programming language. It is designed for the use of data scientists, statisticians, data miners, business intelligence developers. App is designed for the use by data science project teams. Software is powerful and popular IDE for R language. Application is cross-platform product, which is available for Windows, macOS, Ubuntu. Product RStudio app has a simple and intuitive interface that is easy to learn and makes it a good place to start for beginners. Interface has a lot of functionality, and it is great tool for analysing and visualizing data. Application provides a lot of support in form of a built-in help system.

3.1INTERFACE

Software RStudio download is well-designed, intuitive, user-friendly application. Design is minimalistic and clean. Interface is clean and user can easily navigate through the menus. Application provides a wide range of features that are mainly categorized into Data Science, Visualization, Administration. Interface is very user-friendly and has a clean design. It is based on "scratchpad" principle, where the user can create their own projects, or start with

one of many templates that are available. Application provides code completion, which makes it easier to write code.

USABILITY

Application download RStudio with an intuitive, user-friendly interface. Design is clean and minimalistic. Product is simple to use. Application is easy to use and has a simple and intuitive interface. There are many features that make application usable, such as code completion and built-in help system. Is very user- friendly and intuitive application. It is designed for data analysts and statisticians and provides a lot of features, functionality for them. Software install RStudio is easy to install has a quick start guide that is easy to follow. It has a built-in help guide that has short tutorials to get you started.

3.2 FUNCTIONALITY

A wide range of features are available in RStudio install. Features are divided into Data Science, Visualization, Administration. These are main areas of software. Functionality of application is excellent. It has a wide range of features for analysing, visualizing data. It can be used for different purposes, such as data science, web development, other fields. Product RStudio Mac offers a lot of functionality. You can use console to run scripts and use interactive code to explore data sets. Editor and help system are very helpful and informative. Plots and charts you create with app are easy to customize and look beautiful. You can use download RStudio for windows to manage your packages and collaborate with other users.

Is very powerful IDE for R. It has a lot of features that you can use for more comfortable using of application. You can use it open up a project, which is folder with a collection of related documents that make up a complete work session. You can use it to open up a file, which is single document or file with a collection of related data or text.

3.3 SUPPORT

There is wide range of documentation and tutorials available on internet. Application RStudio Windows has a forum where user can ask questions and get an answer. There is online training available. Has a lot of support and you can find a lot of tutorials and how-to

guides on their website on YouTube. You can contact the customer service with any problem that you might have. They have a very active community of developers and users.

4.PACKAGES IN R PROGRAMMING

The package is an appropriate way to organize the work and share it with others. Typically, a package will include code (not only R code!), documentation for the package and the functions inside, some tests to check everything works as it should, and data sets.

Packages in R Programming language are a set of R functions, compiled code, and sample data. These are stored under a directory called “library” within the R environment. By default, R installs a group of packages during installation. Once we start the R console, only the default packages are available by default. Other packages that are already installed need to be loaded explicitly to be utilized by the R program that’s getting to use them.

4.1 FUNCTION USED

glm()

Fitting Generalized Linear Models

Description

glm is used to fit generalized linear models, specified by giving a symbolic description of the linear predictor and a description of the error distribution.

Usage

```
glm(formula, family = gaussian, data, weights, subset,  
    na.action, start = NULL, etastart, mustart, offset,  
    control = list(...), model = TRUE, method = "glm.fit",  
    x = FALSE, y = TRUE, singular.ok = TRUE, contrasts = NULL, ...)
```

```
glm.fit(x, y, weights = rep.int(1, nobs),  
        start = NULL, etastart = NULL, mustart = NULL,  
        offset = rep.int(0, nobs), family = gaussian(),  
        control = list(), intercept = TRUE, singular.ok = TRUE)
```

```
## S3 method for class 'glm'  
weights(object, type = c("prior", "working"), ...)
```

Arguments

formula	an object of class " formula " (or one that can be coerced to that class): a symbolic description of the model to be fitted. The details of model specification are given under ‘Details’.
family	a description of the error distribution and link function to be used in the model. For glm this can be a character string naming a family function, a family function or the result of a call to a family function. For glm.fit only the third option is supported. (See family for details of family functions.)
data	an optional data frame, list or environment (or object coercible by as.data.frame to a data frame) containing the variables in the model. If not found in data, the variables are taken from environment(formula), typically the environment from which glm is called.
weights	an optional vector of ‘prior weights’ to be used in the fitting process. Should be NULL or a numeric vector.
subset	an optional vector specifying a subset of observations to be used in the fitting process.
na.action	a function which indicates what should happen when the data contain NAs. The default is set by the na.action setting of options , and is na.fail if that is unset. The ‘factory-fresh’ default is na.omit . Another possible value is NULL, no action. Value na.exclude can be useful.
start	starting values for the parameters in the linear predictor.
etastart	starting values for the linear predictor.
mustart	starting values for the vector of means.
offset	this can be used to specify an <i>a priori</i> known component to be included in the linear predictor during fitting. This should be NULL or a numeric vector of length equal to the number of cases. One or more offset terms can be included in the formula instead or as well, and if more than one is specified their sum is used. See model.offset .
control	a list of parameters for controlling the fitting process. For glm.fit this is passed to glm.control .
model	a logical value indicating whether <i>model frame</i> should be included as a component of the returned value.
method	the method to be used in fitting the model. The default method "glm.fit" uses iteratively reweighted least squares (IWLS): the alternative "model.frame" returns the model frame and does no fitting. User-supplied fitting functions can be supplied either as a function or a character string naming a function, with a function which takes the same arguments as glm.fit. If specified as a character string it is looked up from within the stats namespace.

x, y	For glm: logical values indicating whether the response vector and model matrix used in the fitting process should be returned as components of the returned value. For glm.fit: x is a design matrix of dimension $n * p$, and y is a vector of observations of length n.
singular.ok	logical; if FALSE a singular fit is an error.
contrasts	an optional list. See the contrasts.arg of model.matrix.default.
intercept	logical. Should an intercept be included in the <i>null</i> model?
object	an object inheriting from class "glm".
type	character, partial matching allowed. Type of weights to extract from the fitted model object. Can be abbreviated.
...	For glm: arguments to be used to form the default control argument if it is not supplied directly. For weights: further arguments passed to or from other methods.

CHAPTER-4

PREDICTION OF LUNG CANCER USING LOGISTIC REGRESSION ANALYSIS

1. Data Source and Description

1.1. Data Source

The dataset for this study is sourced from a publicly available clinical database ‘Kaggle’. The dataset was selected based on its relevance to lung cancer prediction, availability of key variables, and ethical considerations (e.g., de-identified data to protect patient privacy). The use of secondary data ensures that the study adheres to ethical standards without requiring direct patient interaction or institutional review board (IRB) approval for data collection.

1.2. Variables

The dataset comprises a diverse set of demographic and clinical variables identified in the literature as significant predictors of lung cancer risk. These variables provide a comprehensive representation of factors influencing lung cancer diagnosis, enabling the model to capture complex relationships. The variables include:

Demographic

- **Age (in years):** A continuous variable representing the patient’s age at the time of data collection. Age is a known risk factor for lung cancer, with incidence increasing in older populations.
- **Gender (Male/Female):** A categorical variable indicating the patient’s gender. Gender may influence lung cancer risk due to differences in smoking prevalence or biological factors.

Behavioural

- **Smoking History (Never, Former, Current):** A categorical variable describing the patient’s smoking status. Smoking is the leading risk factor for lung cancer, and this variable captures the intensity and duration of exposure.

Medical(history)

- **Shortness of Breath (Yes/No):** A binary variable indicating the presence of respiratory symptoms, which may be associated with lung cancer or other respiratory conditions.

- Chronic Cough (Yes/No): A binary variable representing persistent coughing, a common symptom in lung cancer patients.
- Wheezing (Yes/No): A binary variable indicating respiratory obstruction, which may signal underlying lung pathology.
- Chest Pain (Yes/No): A binary variable that may reflect respiratory or cardiovascular issues, potentially linked to lung cancer.
- Clubbing of Fingernails (Yes/No): A binary clinical sign associated with chronic respiratory or cardiovascular conditions, including lung cancer.
- Lung Cancer Diagnosis (Target: Yes/No): The binary outcome variable, where “Yes” indicates a confirmed lung cancer diagnosis, and “No” indicates the absence of lung cancer.

2. Data Preprocessing

Data preprocessing is a critical step to ensure the quality, reliability, and suitability of the dataset for logistic regression modeling. Poor data quality, such as missing values or outliers, can bias model results and reduce predictive accuracy. The following preprocessing steps were applied systematically.

2.1. Data Cleaning

Missing Values: Missing data was identified and handled using appropriate techniques, such as deletion or imputation, depending on the extent and nature of the missing data. For minimal missingness, listwise deletion was considered. For significant missingness, imputation techniques, such as mean or median imputation for numerical variables or mode imputation for categorical variables, were applied.

Outliers: Outliers in numerical variables (e.g., age) were detected using statistical methods like the Interquartile Range (IQR) method. Values beyond a certain threshold were flagged and either capped or transformed to reduce their impact.

Duplicates: Duplicate records were identified and removed to prevent overfitting and bias.

2.2. Feature Encoding

Categorical variables, such as smoking history and gender, were encoded numerically to make them compatible with logistic regression. One-hot encoding was used for nominal variables with more than two categories (e.g., smoking history: Never, Former, Current), creating binary dummy variables. Binary categorical variables (e.g., gender) were label-encoded (e.g., Male = 0, Female = 1).

3. Load the Data

```
data=read.csv("LUNGDATA1.csv",header=T)
head(data)
>lcfit=glm(LC~GEN+AGE+SMOK+Y.FIGURE+PRESSURE+C.DISEASE+FATIGUE+
ALLERGY+COUGH+S.DIFFICULTY,family=binomial,data=data)
> summary(lcfit)
```

OUTPUT:

Call:

```
glm(formula = LC ~ GEN + AGE + SMOK + Y.FIGURE + PRESSURE + C.DISEASE +
  FATIGUE + ALLERGY + COUGH + S.BREATH, family = binomial,
  data = data)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-6.53386	1.84366	-3.544	0.000394	***
GEN	-1.20339	0.44987	-2.675	0.007473	**
AGE	-0.04331	0.02492	-1.738	0.082245	.
SMOK	1.20575	0.46253	2.607	0.009137	**
Y.FIGURE	1.28672	0.46040	2.795	0.005194	**
PRESSURE	0.91969	0.41444	2.219	0.026478	*
C.DISEASE	2.86508	0.55164	5.194	2.06e-07	***
FATIGUE	2.80350	0.48317	5.802	6.54e-09	***
ALLERGY	1.79795	0.42300	4.250	2.13e-05	***


```
COUGH      2.68739  0.54513  4.930 8.23e-07 ***
S.BREATH   2.24117  0.55407  4.045 5.23e-05 ***
```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 352.60 on 318 degrees of freedom

Residual deviance: 179.73 on 308 degrees of freedom

AIC: 201.73

Number of Fisher Scoring iterations: 7

4.Load the test Data

```
# read the CSV
```

```
> test_data=read.csv("TEST_DATA1.csv",header=T)
```

5. Make Predictions on New Data

To predict probabilities on new data, use the predict() function with type = "response":

```
#predict probabilities on the test data
```

```
predicted_probs=predict(lcfit, newdata = test_data, type = "response")
```

6.Convert Probabilities to Class Labels

To convert predicted probabilities into binary class labels (e.g., 0 or 1), apply a threshold (commonly 0.5):

```
#convert probabilities to class labels using 0.5 thresholds
```

```
> predicted_classes = ifelse(predicted_probs > 0.5, 1, 0)
```

```
#actual values from the test data
```

```
> actual_classes = as.numeric(as.character(test_data$LC))
```

```
#create comparison table
```

```
> comparison =data.frame( Actual = actual_classes,Predicted =  
predicted_classes,Probability = round(predicted_probs, 3) )
```

```
> head(comparison)
```

OUTPUT:

Actual Predicted Probability

1	0	0	0.009
2	0	0	0.004
3	0	0	0.009
4	0	0	0.031
5	0	0	0.467
6	0	0	0.035

7.Calculate accuracy

```
> accuracy <- mean(comparison$Actual == comparison$Predicted)
> cat("Accuracy on test data:", round(accuracy * 100, 2), "%\n")
```

OUTPUT:

Accuracy on test data: 85.37 %

8.Evaluate Model Performance

Compare the predicted classes to the actual classes in your test data to assess model accuracy. For instance, you can create a confusion matrix:

```
table(Actual = comparison$Actual, Predicted = comparison$Predicted)
```

OUTPUT:

Predicted

Actual 0 1

0 49 5

1 7 21

9.INTERPRETATION:

In logistic regression, coefficients represent the change in the log-odds of the outcome for a one-unit increase in the predictor variable, holding other variables constant.

Exponentiating these coefficients yields the odds ratios, which are more interpretable.

Variable	Estimate	Odds Ratio	Significance	Interpretation
(Intercept)	-6.534	—	***	Baseline log-odds of lung cancer when all predictors are 0.
GEN	-1.203	0.30	**	Being male (assuming male=1) reduces odds of lung cancer by 70%.
SMOK	1.206	3.34	**	Smokers have over 3 times higher odds of lung cancer.
Y.FINGURE	1.287	3.62	**	Having yellow fingers significantly increases the odds of the outcome by about 4 times ($e^{1.287} \approx 3.62$), indicating a strong positive association.
PRESSURE	0.920	2.51	*	Individuals with high blood pressure have approximately 2.51 times higher odds of the outcome compared to those with normal blood pressure
C.DISEASE	2.865	17.55	***	Chronic disease increases odds by over 17 times.
FATIGUE	2.804	16.52	***	Individuals experiencing fatigue have 16.52 times higher odds of the outcome compared to those without fatigue

Variable	Estimate	Odds Ratio	Significance	Interpretation
ALLERGY	1.798	6.04	***	Individuals with allergies have 6.04 times higher odds of the outcome compared to those without allergies.
COUGH	2.687	14.70	***	Individuals with a cough have 14.70 times higher odds of the outcome compared to those without a cough.
S.BREATH	2.241	9.40	***	Individuals with Shortness of breath have 9.40 times higher odds of the outcome compared to those without this difficulty

*Significance codes: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, . $p < 0.1$*

The high odds ratios for variables like chronic disease, fatigue, cough, and shortness of breath highlight their strong association with lung cancer. These findings are consistent with known risk factors and symptoms associated with the disease.

Note: The odds ratio is calculated as $\exp(\text{Estimate})$. An odds ratio greater than 1 indicates increased odds of the outcome with the predictor, while an odds ratio less than 1 indicates decreased odds.

✓ Final Remarks

- **Model Strength:** The significant reduction in deviance and low AIC value indicate a strong model fit.
- **Key Predictors:** Chronic disease, fatigue, cough, and shortness of breath are the most influential predictors, substantially increasing the odds of lung cancer.

APPENDIX-A

PATIENT QUESTIONNAIRE:

Section A: Personal Information

1. Gender

☐ Male ☐ Female

2. Age

_____ years old

Section B: Lifestyle & Behavioral Factors

3. Do you smoke or have you ever smoked? (SMOKING)

☐ Yes ☐ No

4. Do you have yellow stains on your fingers? (YELLOW_FINGERS)

☐ Yes ☐ No

5. Do you feel anxious or have been diagnosed with anxiety? (ANXIETY)

☐ Yes ☐ No

**6. Have you ever experienced peer pressure to smoke or drink?
(PEER_PRESSURE)**

☐ Yes ☐ No

7. Do you consume alcohol? (ALCOHOL CONSUMING)

☐ Yes ☐ No

Section C: Medical History

8. Do you suffer from any chronic diseases? (CHRONIC DISEASE)

☐ Yes ☐ No

9. Do you have any allergies? (ALLERGY)

☐ Yes ☐ No

If yes, please specify: _____

Section D: Symptoms (In the Past Month)

10. Have you felt excessive fatigue or tiredness? (FATIGUE)

☐ Yes ☐ No

11. Have you experienced wheezing during breathing? (WHEEZING)

☐ Yes ☐ No

12. Have you been coughing regularly? (COUGHING)

☐ Yes ☐ No

13. Do you have shortness of breath? (SHORTNESS OF BREATH)

☐ Yes ☐ No

14. Do you have difficulty swallowing? (SWALLOWING DIFFICULTY)

☐ Yes ☐ No

15. Do you feel chest pain (even mild or occasional)? (CHEST PAIN)

☐ Yes ☐ NO

APPENDIX-B

The Lung Cancer data used for estimation of Lung Cancer predictive model using logistic regression																
Patient_Id	Gender	Age	SMOKE	Y_FINGURE	ANXIETY	P_PRESSURE	C_DISEASE	FATIGUE	ALLERGY	WHEEZING	ALCOHOL	COUGHING	S_BREATH	S_DIFFICULTY	CHEST PAIN	LC_RISK
P001	MALE	69	1	0	0	1	1	0	1	0	0	0	0	0	0	1
P002	MALE	74	0	1	1	1	0	0	0	1	1	1	0	0	0	1
P003	FEMALE	59	1	1	1	0	1	0	1	0	1	0	0	1	0	0
P004	MALE	63	0	0	0	1	1	1	1	1	0	1	1	0	0	0
P005	FEMALE	63	1	0	1	1	1	1	1	0	1	0	0	1	1	0
P006	FEMALE	75	1	0	1	1	0	0	0	0	1	0	0	1	1	1
P007	MALE	52	0	1	1	1	1	0	1	0	0	0	0	1	0	1
P008	FEMALE	51	0	0	0	0	1	0	0	1	1	1	0	0	1	1
P009	FEMALE	68	0	1	0	1	1	0	1	1	1	1	1	1	1	0
P010	MALE	53	0	0	0	0	0	1	0	1	0	1	1	0	0	1
P011	FEMALE	61	0	0	0	0	0	0	1	0	1	0	0	0	1	1
P012	MALE	72	1	1	1	1	0	0	0	0	0	0	0	1	0	1
P013	FEMALE	60	0	1	1	1	1	0	1	1	1	1	0	1	1	0
P014	MALE	58	0	1	1	1	1	0	0	0	0	0	0	1	0	1
P015	MALE	69	0	1	1	1	1	1	0	0	0	0	1	1	0	0
P016	FEMALE	48	1	0	0	0	0	0	0	0	1	0	0	0	1	1
P017	MALE	75	0	1	1	1	0	1	0	0	0	0	0	1	0	1
P018	MALE	57	0	0	0	0	0	1	1	1	0	1	1	0	0	1
P019	FEMALE	68	0	0	0	0	0	0	1	1	1	0	0	1	1	1
P020	FEMALE	61	1	1	1	1	0	0	1	1	1	1	0	1	1	0
P021	FEMALE	44	0	0	0	0	0	0	1	1	1	1	0	0	1	1
P022	FEMALE	64	1	0	0	0	1	1	0	0	1	0	1	0	1	1
P023	FEMALE	21	0	1	1	1	0	0	0	1	1	1	0	1	1	0
P024	MALE	60	0	1	1	1	1	0	0	0	0	0	0	1	0	1
P025	MALE	72	0	0	0	0	0	1	0	0	0	0	1	0	0	1
P026	MALE	65	1	0	0	1	1	0	1	0	0	0	0	0	0	1
P027	FEMALE	61	0	0	0	1	1	0	0	1	0	1	0	0	0	1
P028	MALE	69	1	1	1	0	1	0	1	0	1	0	0	1	0	0
P029	FEMALE	53	0	0	0	1	0	1	1	0	0	1	0	0	0	1
P030	MALE	55	1	0	1	1	1	0	1	0	0	0	0	1	1	0
P031	FEMALE	57	0	0	1	1	1	1	1	1	1	1	0	1	1	0
P032	MALE	62	0	1	0	1	1	1	0	0	0	1	0	0	0	1
P033	MALE	56	0	0	0	1	1	1	1	1	1	1	0	0	1	0
P034	FEMALE	67	0	0	0	1	0	1	1	1	1	1	0	0	0	1
P035	MALE	59	1	0	0	1	1	1	1	1	1	1	1	0	0	0
P036	FEMALE	59	0	0	0	1	0	1	1	1	1	1	0	0	1	1
P037	MALE	60	1	0	1	1	0	1	1	0	1	0	0	1	0	1
P038	FEMALE	56	1	1	1	1	0	1	1	0	1	1	0	0	1	0
P039	MALE	60	0	1	1	1	0	1	0	0	0	0	1	1	0	1

The Lung Cancer data used for estimation of Lung Cancer predictive model using logistic regression																
Patient_Id	Gender	Age	SMOKEING	Y_FINGURE	ANXIETY	P_PRESSURE	C_DISEASE	FATIGUE	ALLERGY	WHEEZING	ALCOHOL	COUGHING	S_BREATH	S_DIFFICULTY	CHEST_PAIN	LC_RISK
P040	MALE	68	0	1	0	1	1	0	0	1	0	0	0	1	0	1
P041	MALE	63	1	1	1	0	1	0	0	0	0	1	1	0	1	1
P042	FEMALE	77	1	0	0	0	0	0	1	0	0	1	1	1	1	1
P043	FEMALE	52	1	1	1	0	1	0	1	0	1	0	0	1	0	0
P044	MALE	70	0	0	0	1	1	1	1	1	0	1	1	0	0	0
P045	FEMALE	72	1	0	1	1	1	1	1	0	1	0	0	1	1	0
P046	MALE	62	0	1	1	0	1	0	0	0	0	1	0	1	0	1
P047	FEMALE	64	0	0	1	0	0	1	1	1	0	0	1	0	1	1
P048	MALE	70	0	0	0	0	0	0	1	0	0	0	0	0	0	1
P049	MALE	60	0	0	1	1	0	1	0	1	1	0	0	0	0	1
P051	FEMALE	56	0	0	1	0	1	0	1	0	0	0	1	0	0	1
P052	FEMALE	63	1	1	0	0	0	0	0	0	0	1	0	0	0	1
P053	MALE	54	1	1	0	0	0	1	1	1	0	1	1	1	1	0
P054	FEMALE	49	1	1	1	0	0	0	0	0	0	1	1	1	0	1
P055	MALE	57	0	0	0	1	0	0	0	0	1	1	0	1	1	1
P056	FEMALE	52	0	1	1	0	1	0	0	0	0	0	1	0	0	1
P057	MALE	63	0	1	1	0	0	0	0	0	0	0	0	0	0	1
P058	FEMALE	73	1	0	1	0	0	0	0	1	0	0	1	1	1	1
P059	MALE	47	1	0	0	1	0	1	0	0	0	0	1	0	1	0
P060	FEMALE	69	1	0	1	0	1	0	1	1	1	0	0	1	0	1
P061	MALE	70	1	1	1	1	0	1	0	1	0	0	0	0	0	1
P062	MALE	60	1	0	1	0	0	0	1	0	1	1	0	0	0	1
P063	MALE	70	0	0	0	0	1	0	0	1	0	0	0	1	0	1
P064	MALE	68	1	0	1	0	0	0	0	0	0	0	1	0	0	1
P065	FEMALE	74	1	0	0	1	1	1	1	1	1	0	1	1	1	0
P066	MALE	71	1	0	1	0	1	0	0	0	0	0	1	1	1	1
P067	FEMALE	56	1	1	0	1	0	1	0	0	0	1	1	0	1	1
P068	MALE	66	1	0	1	0	1	0	0	0	0	0	0	1	0	1
P069	FEMALE	76	1	1	1	0	1	0	1	0	1	0	0	1	0	0
P070	MALE	78	0	0	0	1	1	1	1	1	0	1	1	0	0	0
P071	FEMALE	68	1	0	1	1	1	1	1	0	1	0	0	1	1	0
P072	FEMALE	66	0	0	0	0	0	0	1	0	1	0	1	0	0	1
P073	FEMALE	67	1	0	1	1	0	0	0	0	1	0	0	1	0	1
P074	MALE	60	0	1	1	1	1	0	1	0	0	0	0	1	1	0
P075	FEMALE	61	0	0	0	0	1	0	0	1	1	1	0	0	0	1
P076	FEMALE	58	0	0	0	0	1	0	1	0	1	0	0	0	1	1
P077	MALE	76	0	0	0	0	1	1	0	1	0	1	1	0	0	1
P078	FEMALE	56	0	0	0	0	1	0	1	0	1	0	0	0	1	1
P079	MALE	67	1	1	1	1	0	0	0	0	0	0	0	1	0	1
P080	FEMALE	73	0	1	1	1	0	0	1	1	1	1	0	1	1	1
P081	MALE	58	0	1	1	1	1	0	0	0	0	0	0	1	0	1

The Lung Cancer data used for estimation of Lung Cancer predictive model using logistic regression																
Patient_Id	Gender	Age	SMOKE	Y_FINGURE	ANXIETY	P_PRESSURE	C_DISEASE	FATIGUE	ALLERGY	WHEEZING	ALCOHOL	COUGHING	S_BREATH	S_DIFFICULTY	CHEST PAIN	LC_RISK
P082	MALE	54	0	1	1	1	1	1	0	0	0	0	1	1	1	1
P083	FEMALE	62	1	0	0	0	0	0	0	0	1	0	0	0	0	1
P084	MALE	81	0	1	1	1	1	0	0	0	0	0	0	1	0	1
P085	MALE	56	0	0	0	0	0	1	1	1	0	1	1	0	0	1
P086	FEMALE	60	0	0	0	0	1	0	1	1	1	0	0	0	0	1
P087	FEMALE	66	1	1	1	1	1	0	1	1	1	1	0	1	1	0
P088	FEMALE	62	0	0	0	0	0	0	1	1	1	1	0	0	1	1
P089	FEMALE	62	0	0	0	0	0	1	0	0	0	1	1	0	0	1
P090	FEMALE	55	1	1	1	0	0	1	0	1	0	0	0	1	1	1
P091	MALE	62	1	1	1	1	0	0	0	1	0	0	0	1	0	1
P092	MALE	71	1	0	0	1	1	1	1	0	0	0	0	0	1	1
P093	MALE	52	1	0	0	1	0	1	0	1	0	0	0	1	0	1
P094	MALE	59	1	0	0	1	1	0	1	0	1	1	1	0	0	1
P095	FEMALE	48	0	0	0	1	0	1	0	1	0	1	1	1	1	1
P096	FEMALE	60	0	1	1	0	0	0	0	0	0	1	1	0	0	1
P097	FEMALE	61	1	1	1	0	1	1	1	0	0	1	1	0	0	1
P098	FEMALE	59	1	1	1	1	0	0	0	1	1	0	0	1	0	1
P099	MALE	64	0	1	1	1	0	0	0	0	0	1	1	0	0	1
P100	FEMALE	56	1	0	0	0	0	1	0	0	0	0	0	0	1	1
P101	MALE	58	0	1	1	1	0	0	0	1	0	0	0	0	0	1
P102	MALE	81	1	0	0	0	1	0	1	1	1	1	1	0	0	1
P103	FEMALE	64	0	0	0	1	1	1	0	1	0	0	0	1	0	1
P104	MALE	62	0	1	1	0	1	1	1	1	0	0	0	1	1	1
P105	MALE	72	1	0	0	0	1	0	0	1	1	0	1	0	1	1
P106	MALE	60	0	1	1	1	1	0	0	0	0	0	0	1	0	1
P107	MALE	61	0	1	1	1	1	1	0	0	0	0	1	1	1	1
P108	FEMALE	60	1	0	0	0	0	0	0	0	1	0	0	0	0	1
P109	MALE	49	0	1	1	1	1	0	0	0	0	0	0	1	0	1
P110	MALE	53	0	0	0	0	0	1	1	1	0	1	1	0	0	1
P111	FEMALE	58	0	0	0	0	1	0	1	1	1	0	0	0	0	1
P112	FEMALE	61	1	1	1	1	0	0	1	1	1	1	0	1	1	1
P113	FEMALE	68	0	0	0	0	0	0	1	1	1	1	0	0	1	1
P114	FEMALE	60	0	0	0	0	0	1	0	0	0	1	1	0	0	1
P115	FEMALE	72	1	1	1	0	0	1	0	1	0	0	0	1	1	0
P116	MALE	72	1	1	1	1	0	0	0	1	0	1	0	1	0	1
P117	MALE	57	1	0	0	1	1	0	1	0	0	0	0	0	0	1
P118	MALE	51	0	0	0	1	1	0	0	1	0	1	0	0	0	1
P119	FEMALE	54	1	1	1	0	1	0	1	0	1	0	0	1	0	1
P120	MALE	56	0	0	0	1	1	1	1	1	0	1	1	0	0	0
P121	FEMALE	77	1	0	1	1	1	0	1	0	0	0	0	1	1	1
P122	FEMALE	64	1	0	1	1	0	0	0	0	1	0	0	1	1	1

The Lung Cancer data used for estimation of Lung Cancer predictive model using logistic regression																
Patient_Id	Gender	Age	SMOKE	Y_FINGURE	ANXIETY	P_PRESSURE	C_DISEASE	FATIGUE	ALLERGY	WHEEZING	ALCOHOL	COUGHING	S_BREATH	S_DIFFICULTY	CHEST PAIN	LC_RISK
P123	MALE	57	0	1	1	1	1	0	1	0	0	0	0	1	0	1
P124	FEMALE	66	0	0	0	0	1	0	0	1	1	1	0	0	1	0
P125	FEMALE	70	0	0	0	0	1	0	1	0	1	0	0	0	1	1
P126	FEMALE	53	1	0	0	0	1	1	0	1	0	1	1	0	0	0
P127	MALE	51	0	0	0	0	1	0	1	0	1	0	0	0	1	1
P128	MALE	58	1	1	1	1	0	0	0	0	0	0	0	1	0	1
P129	MALE	58	0	1	0	1	0	0	1	1	1	1	0	1	1	1
P130	FEMALE	63	0	0	0	1	0	0	0	0	0	0	0	1	1	1
P131	MALE	51	0	1	1	1	1	1	0	1	0	0	1	1	0	0
P132	FEMALE	61	1	0	0	0	0	0	0	1	1	0	0	1	1	1
P133	MALE	61	0	1	1	1	1	0	1	0	0	0	0	1	0	1
P134	MALE	76	0	0	0	0	0	1	1	1	0	1	1	0	0	1
P135	FEMALE	71	0	0	0	0	1	0	1	1	1	0	0	0	1	1
P136	FEMALE	69	1	1	1	1	0	0	1	1	1	1	0	1	1	0
P137	FEMALE	56	0	0	0	0	1	0	1	1	1	1	0	0	1	1
P138	FEMALE	67	1	0	0	0	1	1	0	0	1	0	1	0	1	1
P139	FEMALE	54	0	1	1	1	0	0	0	1	1	1	0	1	1	1
P140	MALE	63	0	1	1	1	1	0	0	0	0	0	0	1	0	1
P141	MALE	47	0	0	0	1	0	1	0	0	0	0	1	0	0	1
P142	MALE	62	1	1	0	1	1	0	1	0	0	0	0	0	1	1
P143	FEMALE	65	0	0	0	1	1	0	0	1	1	1	0	1	0	1
P144	MALE	63	1	1	1	0	1	0	1	0	1	0	0	1	0	1
P145	FEMALE	64	0	0	0	1	0	1	1	0	0	1	0	0	0	1
P146	FEMALE	65	1	1	1	0	1	0	1	0	1	0	0	1	0	0
P147	MALE	51	0	0	0	1	1	1	1	1	0	1	1	0	0	0
P148	FEMALE	56	1	0	1	1	1	1	1	0	1	0	0	1	1	0
P149	MALE	70	1	0	1	1	1	0	1	0	0	0	0	1	1	1
P150	FEMALE	58	0	0	1	0	0	0	0	0	1	0	0	1	1	1
P151	MALE	67	0	1	0	1	1	0	1	0	0	0	0	1	0	1
P152	MALE	62	0	0	0	0	1	0	0	1	1	1	0	0	1	1
P153	FEMALE	74	0	0	0	0	0	0	0	0	1	0	0	0	0	1
P154	MALE	69	1	0	0	0	1	1	0	1	0	1	1	0	0	1
P155	FEMALE	64	0	0	0	0	1	0	1	0	1	0	0	0	1	1
P156	MALE	75	1	0	1	1	0	0	0	0	0	0	0	1	0	1
P157	FEMALE	47	1	1	1	0	0	0	1	1	0	0	0	0	1	1
P158	MALE	57	0	1	1	1	1	0	0	0	0	0	0	1	0	1
P159	MALE	56	0	1	1	1	1	1	0	0	0	0	1	1	0	1
P160	MALE	68	0	1	0	1	1	0	0	1	0	0	0	1	0	1
P161	MALE	55	1	1	1	0	1	0	0	0	0	1	1	0	1	1
P162	FEMALE	62	1	0	0	0	0	0	1	0	0	1	1	1	1	1
P163	MALE	73	0	1	1	0	1	1	1	1	1	1	1	1	0	0

The Lung Cancer data used for estimation of Lung Cancer predictive model using logistic regression																
Patient_Id	Gender	Age	SMOKE	Y_FINGURE	ANXIETY	P_PRESSURE	C_DISEASE	FATIGUE	ALLERGY	WHEEZING	ALCOHOL	COUGHING	S_BREATH	S_DIFFICULTY	CHEST PAIN	LC_RISK
P164	FEMALE	68	0	0	1	0	0	1	1	1	1	1	1	1	1	0
P165	FEMALE	75	1	1	1	0	1	0	1	0	1	0	0	1	0	0
P166	MALE	63	0	0	0	1	1	1	1	1	0	1	1	0	0	0
P167	FEMALE	61	1	0	1	1	1	1	1	0	1	0	0	1	1	0
P168	MALE	62	0	0	0	0	0	1	1	1	1	1	1	1	0	1
P169	MALE	44	0	0	1	1	0	1	1	1	1	1	1	1	0	0
P170	FEMALE	56	0	0	1	0	1	1	1	1	1	1	1	1	0	0
P171	FEMALE	54	1	1	0	0	0	0	0	0	0	1	0	0	0	1
P172	MALE	57	1	1	0	0	0	1	1	1	0	1	1	1	1	0
P173	FEMALE	56	1	1	1	0	0	0	0	0	0	1	1	1	0	1
P174	MALE	69	0	0	0	1	0	0	0	0	1	1	0	1	1	1
P175	FEMALE	72	0	1	1	0	1	0	0	0	0	0	1	0	0	1
P176	MALE	59	0	1	1	0	0	0	0	0	0	0	0	0	0	1
P177	FEMALE	70	1	0	1	0	0	0	0	1	0	0	1	1	1	1
P178	MALE	64	1	0	0	1	0	1	0	0	0	0	1	0	1	1
P179	FEMALE	61	1	0	1	0	1	0	1	1	1	0	0	1	0	1
P180	MALE	72	1	1	1	1	0	1	0	1	0	0	0	0	0	1
P181	MALE	63	1	0	1	0	0	0	1	0	1	1	0	0	0	1
P182	MALE	74	0	0	0	0	1	0	0	1	0	0	0	1	0	1
P183	MALE	71	1	0	1	0	0	0	0	0	0	0	1	0	0	1
P184	FEMALE	71	1	0	0	1	1	1	1	1	1	0	1	1	1	0
P185	MALE	72	1	0	1	0	1	0	0	0	0	0	1	1	1	1
P186	FEMALE	77	1	1	0	1	0	1	0	0	0	1	1	0	1	0
P187	MALE	72	1	0	1	0	1	0	0	0	0	0	0	1	0	1
P188	FEMALE	55	0	0	0	0	0	0	1	0	1	0	1	0	0	1
P189	FEMALE	65	1	0	1	1	0	0	0	0	1	0	0	1	0	1
P190	MALE	67	0	1	1	0	1	1	1	1	1	1	1	1	0	0
P191	FEMALE	69	0	0	1	0	0	1	1	1	1	1	1	1	1	0
P192	FEMALE	55	1	1	1	0	1	0	1	0	1	0	0	1	0	0
P193	MALE	51	0	0	0	1	1	1	1	1	0	1	1	0	0	0
P194	FEMALE	64	1	0	1	1	1	1	1	0	1	0	0	1	1	0
P195	MALE	63	1	1	1	1	0	0	0	0	0	0	0	1	0	1
P196	FEMALE	69	0	1	1	1	0	0	1	1	1	1	0	1	1	0
P197	MALE	64	0	1	1	1	1	0	0	0	0	0	0	1	0	1
P198	MALE	59	0	1	1	1	1	1	0	0	0	0	1	1	1	1
P199	FEMALE	73	1	0	0	0	0	0	1	1	1	1	1	1	1	1
P200	MALE	55	0	1	1	1	1	0	1	1	1	1	1	1	1	1
P201	MALE	63	0	0	0	0	0	1	1	1	1	1	1	1	1	1
P202	FEMALE	60	0	0	0	0	1	0	1	1	1	1	1	1	1	0
P203	FEMALE	74	1	1	1	1	0	0	1	1	1	1	1	1	1	1
P204	FEMALE	65	0	0	0	0	0	0	1	1	1	1	1	1	1	1

The Lung Cancer data used for estimation of Lung Cancer predictive model using logistic regression																
Patient_Id	Gender	Age	SMOKE	Y_FINGURE	ANXIETY	P_PRESSURE	C_DISEASE	FATIGUE	ALLERGY	WHEEZING	ALCOHOL	COUGHING	S_BREATH	S_DIFFICULTY	CHEST PAIN	LC_RISK
P205	FEMALE	79	0	0	0	0	0	1	1	1	1	1	1	1	1	0
P206	FEMALE	62	1	1	1	0	0	1	1	1	1	1	1	1	1	1
P207	MALE	71	1	1	1	1	0	0	0	1	0	0	0	1	0	1
P208	MALE	63	1	0	0	1	1	1	1	0	0	0	0	0	1	0
P209	MALE	67	1	0	0	1	0	1	0	1	0	0	0	1	0	1
P210	MALE	55	1	0	0	1	1	0	1	0	1	1	1	0	0	1
P211	FEMALE	54	0	0	0	1	0	1	0	1	0	1	1	1	1	1
P212	FEMALE	77	0	1	1	0	0	0	0	0	0	1	1	0	0	1
P213	FEMALE	58	1	1	1	0	1	1	1	0	0	1	1	0	0	0
P214	FEMALE	64	1	1	1	1	0	0	0	1	1	0	0	1	0	1
P215	MALE	61	0	1	1	1	0	0	0	0	0	1	1	0	0	1
P216	FEMALE	62	1	0	0	0	0	1	0	0	0	0	0	0	1	1
P217	MALE	67	0	1	1	1	0	0	0	1	0	0	0	0	0	1
P218	MALE	56	1	0	0	0	1	0	1	1	1	1	1	0	0	1
P219	FEMALE	70	0	0	0	1	1	1	0	1	0	0	0	1	0	1
P220	MALE	70	0	1	1	0	1	1	1	1	1	0	0	1	1	0
P221	MALE	57	1	0	0	0	1	0	0	1	1	0	1	0	1	1
P222	MALE	61	0	1	1	1	1	0	0	0	0	0	0	1	0	1
P223	MALE	77	0	1	1	1	1	1	0	0	0	0	1	1	1	1
P224	FEMALE	63	1	0	0	0	0	0	0	0	1	0	0	0	0	1
P225	MALE	62	0	1	1	1	1	0	0	0	0	0	0	1	0	1
P226	MALE	59	0	0	0	0	0	1	1	1	0	1	1	0	0	1
P227	FEMALE	70	0	0	0	0	1	0	1	1	1	0	0	0	0	1
P228	FEMALE	71	1	1	1	1	0	0	1	1	1	1	0	1	1	0
P229	FEMALE	56	0	0	0	0	0	0	1	1	1	1	0	0	1	1
P230	FEMALE	57	0	0	0	0	0	1	0	0	0	1	1	0	0	1
P231	FEMALE	78	1	1	1	0	0	1	0	1	0	0	0	1	1	1
P232	MALE	64	1	1	1	1	0	0	0	1	0	0	0	1	0	1
P233	FEMALE	62	1	1	0	0	0	0	0	0	0	1	0	0	0	1
P234	MALE	49	1	1	0	0	0	1	1	1	0	1	1	1	1	0
P235	FEMALE	77	1	1	1	0	0	0	0	0	0	1	1	1	0	1
P236	MALE	64	0	0	0	1	0	0	0	0	1	1	0	1	1	1
P237	FEMALE	63	0	1	1	0	1	0	0	0	0	0	1	0	0	1
P238	MALE	54	0	1	1	0	0	0	0	0	0	0	0	0	0	1
P239	FEMALE	38	1	0	1	0	0	0	0	1	0	0	1	1	1	1
P240	MALE	75	1	0	0	1	0	1	0	0	0	0	1	0	1	1
P241	FEMALE	70	1	0	1	0	1	0	1	1	1	0	0	1	0	1
P242	MALE	59	1	1	1	1	0	1	0	1	0	0	0	0	0	1
P243	MALE	77	1	0	1	0	0	0	1	0	1	1	0	0	0	1
P244	MALE	61	0	0	0	0	1	0	0	1	0	0	0	1	0	1
P245	MALE	64	1	0	1	0	0	0	0	0	0	0	1	0	0	1

The Lung Cancer data used for estimation of Lung Cancer predictive model using logistic regression																
Patient_Id	Gender	Age	SMOKE	Y_FINGURE	ANXIETY	P_PRESSURE	C_DISEASE	FATIGUE	ALLERGY	WHEEZING	ALCOHOL	COUGHING	S_BREATH	S_DIFFICULTY	CHEST PAIN	LC_RISK
P246	FEMALE	59	1	0	0	1	1	1	1	1	1	0	1	1	1	1
P247	MALE	71	1	0	1	0	1	0	0	0	0	0	1	1	1	1
P248	FEMALE	67	1	1	0	1	0	1	0	0	0	1	1	0	1	1
P249	MALE	64	1	0	1	0	1	0	0	0	0	0	0	1	0	1
P250	FEMALE	68	0	0	0	0	0	0	1	0	1	0	1	0	0	1
P251	FEMALE	69	1	0	1	1	0	0	0	0	1	0	0	1	0	1
P252	FEMALE	64	1	0	0	0	1	1	0	0	1	0	1	0	1	1
P253	FEMALE	59	0	1	1	0	0	1	0	1	1	1	0	1	1	1
P254	MALE	67	0	1	1	1	1	0	0	0	0	0	0	1	0	1
P255	MALE	74	0	0	0	1	0	1	0	0	1	1	1	0	0	1
P256	MALE	77	1	1	0	1	0	0	1	0	0	0	0	0	1	1
P257	FEMALE	60	0	0	0	1	1	0	0	1	1	1	0	1	0	1
P258	MALE	64	1	1	1	0	1	0	1	0	1	1	0	1	0	0
P259	FEMALE	70	0	0	0	1	0	1	1	0	0	1	0	0	0	1
P260	MALE	58	1	0	1	1	1	0	1	0	0	0	0	1	1	1
P261	FEMALE	59	0	0	1	0	0	0	0	0	1	0	0	1	1	1
P262	MALE	39	0	1	0	1	1	0	1	1	1	1	1	1	1	0
P263	MALE	67	0	0	0	0	1	0	0	1	1	1	0	0	1	1
P264	FEMALE	71	0	0	0	0	0	0	0	0	1	0	0	0	0	1
P265	MALE	70	1	0	0	0	0	1	0	1	0	1	1	0	0	0
P266	FEMALE	60	0	0	0	0	1	0	1	0	1	0	0	0	1	1
P267	MALE	55	1	0	1	1	0	0	0	0	0	0	0	1	0	1
P268	FEMALE	60	1	1	1	0	0	0	1	1	0	0	0	0	1	1
P269	MALE	55	0	1	1	1	1	0	0	0	0	0	0	1	0	1
P270	MALE	55	0	1	1	1	1	1	0	0	0	0	1	1	0	0
P271	MALE	70	0	1	0	1	1	0	0	1	0	0	0	1	0	1
P272	MALE	63	1	1	1	0	1	0	0	0	0	1	1	0	1	1
P273	FEMALE	64	1	0	0	0	0	0	1	0	0	1	1	1	1	1
P274	MALE	59	0	1	1	0	1	0	0	0	0	1	0	1	0	1
P275	FEMALE	56	1	0	1	1	1	1	1	0	1	0	0	1	1	0
P276	FEMALE	64	1	0	1	1	0	0	0	0	1	0	0	1	1	1
P277	MALE	62	0	1	1	1	1	0	1	0	0	0	0	1	0	1
P278	FEMALE	87	0	0	0	0	1	0	0	1	1	1	0	0	1	1
P279	FEMALE	77	0	1	0	1	1	0	1	1	1	1	1	1	1	0
P280	MALE	59	0	0	0	0	0	1	0	1	0	1	1	0	0	1
P281	FEMALE	59	0	0	0	0	0	0	1	0	1	0	0	0	1	1
P282	MALE	55	1	1	1	1	0	0	0	0	0	0	0	1	0	1
P283	FEMALE	46	0	1	1	1	1	0	1	1	1	1	0	1	1	0
P284	MALE	60	0	1	1	1	1	0	0	0	0	0	0	1	0	1
P285	MALE	58	0	1	1	1	1	1	0	0	0	0	1	1	0	0
P286	FEMALE	58	1	0	0	0	0	0	0	0	1	0	0	0	1	1

The Lung Cancer data used for estimation of Lung Cancer predictive model using logistic regression																
Patient_Id	Gender	Age	SMOKE	Y_FINGURE	ANXIETY	P_PRESSURE	C_DISEASE	FATIGUE	ALLERGY	WHEEZING	ALCOHOL	COUGHING	S_BREATH	S_DIFFICULTY	CHEST PAIN	LC_RISK
P287	MALE	63	0	1	1	1	0	1	0	0	0	0	0	1	0	1
P288	MALE	51	0	0	0	0	0	1	1	1	0	1	1	0	0	1
P289	FEMALE	61	0	0	0	0	0	0	1	1	1	0	0	1	1	1
P290	FEMALE	61	1	1	1	1	0	0	1	1	1	1	0	1	1	0
P291	FEMALE	76	0	0	0	0	0	0	1	1	1	1	0	0	1	1
P292	FEMALE	71	1	0	0	0	1	1	0	0	1	0	1	0	1	1
P293	FEMALE	69	0	1	1	1	0	0	0	1	1	1	0	1	1	0
P294	MALE	56	0	1	1	1	1	0	0	1	1	1	0	1	0	0
P295	MALE	67	1	0	0	1	1	1	1	1	1	1	1	0	0	0
P296	MALE	54	1	0	0	1	1	0	1	0	0	0	0	0	0	1
P297	MALE	63	0	0	0	0	0	1	1	1	0	1	1	0	0	1
P298	FEMALE	47	0	0	0	0	1	0	1	1	1	0	0	0	1	1
P299	FEMALE	62	1	1	1	1	0	0	1	1	1	1	0	1	1	0
P300	FEMALE	65	0	0	0	0	1	0	1	1	1	1	0	0	1	1
P301	FEMALE	63	1	0	0	0	1	1	0	0	1	0	1	0	1	1
P302	FEMALE	64	0	1	1	1	0	0	0	1	1	1	0	1	1	0
P303	MALE	65	0	1	1	1	1	0	0	0	0	0	0	1	0	1
P304	MALE	51	0	0	0	1	0	1	0	0	0	0	1	0	0	1
P305	MALE	56	1	1	0	1	1	0	1	0	0	0	0	0	1	1
P306	FEMALE	70	0	0	0	1	1	0	0	1	1	1	0	1	0	0
P307	MALE	58	0	1	1	0	1	1	1	1	1	1	1	1	0	0
P308	FEMALE	67	0	0	1	0	0	1	1	1	1	1	1	1	1	0
P309	FEMALE	62	1	1	1	0	1	0	1	0	1	0	0	1	0	0
P310	MALE	59	0	0	0	1	1	1	1	1	0	1	1	0	0	0
P311	FEMALE	63	1	0	1	1	1	1	1	0	1	0	0	1	1	0
P312	MALE	63	1	1	1	0	1	0	1	0	1	0	0	1	0	1
P313	FEMALE	67	0	0	0	1	0	1	1	0	0	1	0	0	0	1
P314	MALE	54	1	0	1	1	1	0	1	0	0	0	0	1	1	1
P315	FEMALE	63	0	0	1	0	0	0	0	0	1	0	0	1	1	1
P316	MALE	47	0	1	0	1	1	0	1	0	0	0	0	1	0	1
P317	MALE	62	0	0	0	0	1	0	0	1	1	1	0	0	1	1
P318	FEMALE	65	0	0	0	0	0	0	0	0	1	0	0	0	0	1
P319	MALE	63	1	0	0	0	1	1	0	1	0	1	1	0	0	1
P320	FEMALE	65	0	0	0	0	1	0	1	0	1	0	0	0	1	1

APPENDIX-C

The Lung Cancer data used for testing the estimated Lung Cancer predictive model																
Patient_Id	Gender	Age	SMOKE	Y_FINGER	ANXIETY	P_PRESSURE	C_DISEASE	FATIGUE	ALLERGY	WHEEZING	ALCOHOL	COUGHING	S_BREATH	S_DIFFICULTY	CHEST PAIN	LC_RISK
P321	MALE	51	1	0	1	1	0	0	0	0	0	0	0	1	0	1
P322	FEMALE	56	1	1	1	0	0	0	1	1	0	0	0	0	1	1
P323	MALE	70	0	1	1	1	1	0	0	0	0	0	0	1	0	1
P324	MALE	58	0	1	1	1	1	1	0	0	0	0	1	1	0	1
P325	MALE	67	0	1	0	1	1	0	0	1	0	0	0	1	0	1
P326	MALE	62	1	1	1	0	1	0	0	0	0	1	1	0	1	1
P327	FEMALE	55	0	0	0	0	0	0	1	1	1	1	1	1	1	1
P328	FEMALE	51	0	0	0	0	0	1	1	1	1	1	1	1	1	1
P329	FEMALE	64	1	1	1	0	0	1	1	1	1	1	1	1	1	1
P330	MALE	63	1	1	1	1	0	0	0	1	0	0	0	1	0	1
P331	MALE	69	1	0	0	1	1	1	1	0	0	0	0	0	1	1
P332	MALE	64	1	0	0	1	0	1	0	1	0	0	0	1	0	0
P333	MALE	59	1	0	0	1	1	0	1	0	1	1	1	0	0	1
P334	FEMALE	73	0	0	0	1	0	1	0	1	0	1	1	1	1	1
P335	FEMALE	55	0	1	1	0	0	0	0	0	0	1	1	0	0	1
P336	FEMALE	63	1	1	1	0	1	1	1	0	0	1	1	0	0	1
P337	FEMALE	56	0	0	0	1	1	0	0	1	1	1	0	1	0	1
P338	MALE	69	0	1	1	0	1	1	1	1	1	1	1	1	0	0
P339	FEMALE	64	0	0	1	0	0	1	1	1	1	1	1	1	1	0
P340	FEMALE	59	1	1	1	0	1	0	1	0	1	0	0	1	0	0
P341	MALE	63	0	0	0	1	1	1	1	1	0	1	1	0	0	0
P342	FEMALE	63	1	0	1	1	1	1	1	0	1	0	0	1	1	0
P343	MALE	67	1	1	1	0	1	0	1	0	1	0	0	1	0	1
P344	FEMALE	54	0	0	0	1	0	1	1	0	0	1	0	0	0	1
P345	MALE	63	1	0	1	1	1	0	1	0	0	0	0	1	1	1
P346	FEMALE	47	0	0	1	0	0	0	0	0	1	0	0	1	1	0
P347	MALE	62	0	1	0	1	1	0	1	0	0	0	0	1	0	1
P348	MALE	65	0	0	0	0	1	0	0	1	1	1	0	0	1	1
P349	FEMALE	63	0	0	0	0	0	0	0	0	1	0	0	0	0	1
P350	MALE	64	1	0	0	0	1	1	0	1	0	1	1	0	0	1
P351	FEMALE	65	0	0	0	0	1	0	1	0	1	0	0	0	1	1
P352	MALE	51	1	0	1	1	0	0	0	0	0	0	0	1	0	1
P353	FEMALE	61	1	1	1	1	0	0	1	1	1	1	0	1	1	0
P354	FEMALE	44	0	0	0	0	0	0	1	1	1	1	0	0	1	1
P355	FEMALE	64	1	0	0	0	1	1	0	0	1	0	1	0	1	1
P356	FEMALE	21	0	1	1	1	0	0	0	1	1	1	0	1	1	0
P357	MALE	60	0	1	1	1	1	0	0	0	0	0	0	1	0	1
P358	MALE	72	0	0	0	0	0	1	0	0	0	0	1	0	0	1
P359	MALE	65	1	0	0	1	1	0	1	0	0	0	0	0	0	1
P360	FEMALE	61	0	0	0	1	1	0	0	1	0	1	0	0	0	1

The Lung Cancer data used for testing the estimated Lung Cancer predictive model																
patient_id	Gender	Age	SMOKE	Y_FINGURE	ANXIETY	P_PRESSURE	C_DISEASE	FATIGUE	ALLERGY	WHEEZING	ALCOHOL	COUGHING	S_BREATH	S_DIFFICULTY	CHEST PAIN	LC_RISK
P361	MALE	69	1	1	1	0	1	0	1	0	1	0	0	1	0	0
P362	FEMALE	53	0	0	0	1	0	1	1	0	0	1	0	0	0	1
P363	MALE	55	1	0	1	1	1	0	1	0	0	0	0	1	1	0
P364	FEMALE	57	0	0	1	1	1	1	1	1	1	1	0	1	1	0
P365	MALE	62	0	1	0	1	1	1	0	0	0	1	0	0	0	1
P366	MALE	56	0	0	0	1	1	1	1	1	1	1	0	0	1	0
P367	FEMALE	67	0	0	0	1	0	1	1	1	1	1	0	0	0	1
P368	MALE	59	1	0	0	1	1	1	1	1	1	1	1	0	0	0
P369	FEMALE	59	0	0	0	1	0	1	1	1	1	1	0	0	1	1
P370	MALE	60	1	0	1	1	0	1	1	0	1	0	0	1	0	1
P371	FEMALE	56	1	1	1	1	0	1	1	0	1	1	0	0	1	0
P372	MALE	56	0	1	1	1	0	1	1	0	1	1	0	1	0	1
P373	MALE	60	0	1	1	1	0	1	0	0	0	0	1	1	0	1
P374	MALE	68	0	1	0	1	1	0	0	1	0	0	0	1	0	1
P375	MALE	63	1	1	1	0	1	0	0	0	0	1	1	0	1	1
P376	FEMALE	77	1	0	0	0	0	0	1	0	0	1	1	1	1	1
P377	FEMALE	59	1	1	1	0	1	0	1	0	1	0	0	1	0	0
P378	MALE	63	0	0	0	1	1	1	1	1	0	1	1	0	0	0
P379	FEMALE	63	1	0	1	1	1	1	1	0	1	0	0	1	1	0
P380	MALE	52	0	1	1	0	1	0	0	0	0	1	0	1	0	1
P381	FEMALE	70	0	0	1	0	0	1	1	1	0	0	1	0	1	1
P382	MALE	72	0	0	0	0	0	0	1	0	0	0	0	0	0	1
P383	MALE	62	0	0	1	1	0	1	0	1	1	0	0	0	0	1
P384	MALE	74	0	1	1	1	0	0	0	1	1	1	0	0	0	1
P385	FEMALE	59	1	1	1	0	1	0	1	0	1	0	0	1	0	0
P386	MALE	63	0	0	0	1	1	1	1	1	0	1	1	0	0	0
P387	FEMALE	63	1	0	1	1	1	1	1	0	1	0	0	1	1	0
P388	FEMALE	75	1	0	1	1	0	0	0	0	1	0	0	1	1	1
P389	MALE	52	0	1	1	1	1	0	1	0	0	0	0	1	0	1
P390	FEMALE	51	0	0	0	0	1	0	0	1	1	1	0	0	1	0
P391	FEMALE	59	0	0	0	0	0	0	1	0	1	0	1	0	0	1
P392	FEMALE	70	1	0	1	1	0	0	0	0	1	0	0	1	0	1
P393	MALE	69	0	1	1	0	1	1	1	1	1	1	1	1	0	0
P394	FEMALE	64	0	0	1	0	0	1	1	1	1	1	1	1	1	0
P395	FEMALE	59	1	1	1	0	1	0	1	0	1	0	0	1	0	0
P396	MALE	63	0	0	0	1	1	1	1	1	0	1	1	0	0	0
P397	FEMALE	63	1	0	1	1	1	1	1	0	1	0	0	1	1	0
P398	MALE	71	1	1	1	1	0	0	0	0	0	0	0	1	0	1
P399	FEMALE	71	0	1	1	1	0	0	1	1	1	1	0	1	1	0

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